

HR ATS

Applicant Tracking System

Project documentation | Built with n8n, Google Sheets, Gmail, Cal.com and OpenRouter

1. Overview

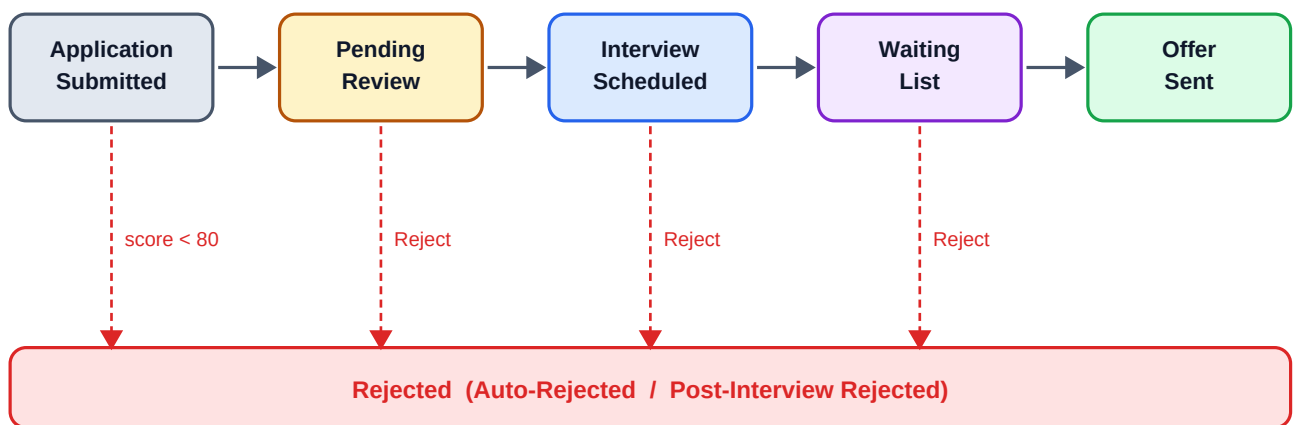
The HR ATS automates the first stages of hiring. Candidates apply through a web form and an AI model scores their CV against the job description. Strong matches are queued for a human decision on a web dashboard, where HR can invite candidates to interview, record interview scores, manage a waiting list and send job offers. Candidate emails are sent automatically at each decision point.

Component	Role
n8n	Runs the four workflows and serves the dashboard through webhooks
Google Sheets	The database: one sheet, "All Applicants", holds every candidate
Gmail	Sends invitations, offers and rejection notices, and the HR daily digest
Cal.com	Lets candidates book their own interview slot; a webhook syncs the booking back
OpenRouter (AI)	Reads the CV and extracts skills, experience, education and a fit summary

2. The four workflows

Workflow	Trigger	Purpose
HR Evaluation	Application form submission	Validate the CV, detect duplicates, score with AI, log to the sheet, auto-reject weak matches
HR Daily Digest	Schedule, every day at 09:00	Email HR the number of pending candidates by role, with a link to the dashboard
HR ATS Web Dashboard	Webhooks (GET)	Show the dashboard and process every HR action (accept, reject, waitlist, offer)
Interview Booking Sync	Cal.com webhook (POST)	Write the interview date and meeting link into the candidate's row

2.1 Candidate lifecycle



2.2 Status values used in the sheet

Status	Meaning	Set by
Pending Review	CV score is 80 or higher; waiting for HR to decide	HR Evaluation
Auto-Rejected	CV score is below 80; rejection email sent automatically	HR Evaluation
Interview Scheduled	HR accepted the CV; invitation with booking link sent	Dashboard
Waiting List	Interviewed and saved for later, ranked by interview score	Dashboard
Offer Sent	Job offer email sent	Dashboard
Post-Interview Rejected	Rejected by HR from the dashboard	Dashboard

3. Workflow details

3.1 HR Evaluation

- **Form:** full name, email, phone, position (Frontend Engineer, Data Analyst or Product Manager), years of experience and a PDF resume (max 5 MB).
- **Validate PDF File:** rejects non-PDF files and files over 5 MB.
- **Check Existing Applicant / Is Duplicate?:** looks for the same email or phone in the sheet. A duplicate gets an "already on file" email and the flow stops.
- **Extract and clean text:** reads the PDF text and trims it to about 12,000 characters.
- **Prepare JD and prompt:** picks the job description and its weighted skills for the chosen position and builds the AI prompt.
- **Stage 1 (AI):** the model returns JSON with skills found, missing skills, years of experience, education and a fit summary.
- **Parse and compute score:** calculates the final score (formula below).
- **Log to sheet and route:** the candidate is appended to the sheet. Score 80+ becomes Pending Review; lower scores get the automatic rejection email.

Scoring formula

Part	Weight	How it is calculated
Required skills	60%	Weighted share of the position's required skills found in the CV
Experience	30%	Detected years divided by the minimum years for the role (capped at 100%)
Education	10%	Full points if the education matches, otherwise 30% of the points
Nice-to-have bonus	up to +10	Share of the nice-to-have skills found, scaled to 10 points

3.2 HR Daily Digest

Every day at 09:00 the workflow reads all rows with status *Pending Review*. If there are any, it groups them by position and emails HR a summary with a button that opens the dashboard.

3.3 HR ATS Web Dashboard

Two webhooks, both under the n8n /webhook/ path:

Path	What it does
ats-dashboard	Reads the sheet and returns the HTML dashboard: summary counters plus three tables (Waiting List, Post-Interview Evaluations, Pending CV Reviews).
ats-action	Receives a button click, updates the candidate's row, sends the matching email, then redirects back to the dashboard.

Actions handled by ats-action

Button (query parameters)	New status	Email sent
Accept for Interview action=approve_cv	Interview Scheduled	Interview invitation with the Cal.com booking link
Reject (pending CV) action=reject_cv	Post-Interview Rejected	Rejection mail
Save to Waitlist action=evaluate_interview decision=waitlist&score=N	Waiting List	None (fixed: previously sent a rejection by mistake)
Reject after interview action=evaluate_interview decision=reject&score=N	Post-Interview Rejected	Rejection mail
Send Offer action=send_offer	Offer Sent	Formal job offer
Reject (waiting list) action=reject_waitlist	Post-Interview Rejected	Rejection mail

Routing inside the workflow: **Update row in sheet** → **If** (approve_cv) → **If Offer** (send_offer) → **If Waitlist** (decision = waitlist) → **Rejection Mail**. Only requests that are none of the first three reach the rejection email.

3.4 Interview Booking Sync

When a candidate books a slot on Cal.com, Cal.com calls the cal -booking webhook. The workflow finds the row by the attendee's email and writes *interviewDate* (converted to Cairo time) and *meetingLink* (the video call URL, or the location if there is none). The dashboard then shows the date and a "Join Google Meet" button.

4. Google Sheet columns (All Applicants)

Column	Filled by	Description
applicantName, applicantEmail, applicantPhone	HR Evaluation	Contact details from the form (email is the key used to find a row)
appliedPosition	HR Evaluation	Position the candidate applied for
extractedSkills, experienceYearsActual, educationDetected	HR Evaluation	Details extracted from the CV by the AI
finalScore, fitSummary	HR Evaluation	CV score (0-100) and the AI's summary of the candidate
status, applicationDate	HR Evaluation / Dashboard	Current pipeline status and the submission timestamp
interviewDate, meetingLink	Interview Booking Sync	Interview time (Cairo) and video link
interviewScore, interviewNotes	Dashboard	HR's interview score (0-100) and optional notes

5. Setup checklist

- Create n8n credentials for **Google Sheets OAuth2** and **Gmail OAuth2**, and select them in every Sheets and Gmail node.
- Store the **OpenRouter API key** as an n8n credential (header auth) instead of typing it into the HTTP Request node.
- Make sure the sheet has a header row with the column names listed above and a tab named *All Applicants*.
- Replace the placeholder URLs: the dashboard link in the Daily Digest code and the `localhost:5678` addresses in the dashboard code must point to your public n8n address.
- In Cal.com, add a webhook for *booking created* that points to `/webhook/cal-booking` (POST).
- Publish (activate) all four workflows. Use the production webhook URLs, not the test URLs.

6. Known issues and recommendations

Issue	Recommendation
API key stored in plain text in the HTTP Request node	Rotate the key and move it into an n8n credential
Candidates scoring 60-79 are auto-rejected although the Switch output is labelled "Manual Review (60-79)"	Change the rules to 60-79 and 80+, and add a status for manual review if that is the intent
Rejecting a pending CV sends the post-interview email wording and saves "Post-Interview Rejected"	Use a separate CV-stage email and the status Auto-Rejected / CV Rejected
Send Offer and Reject (waiting list) overwrite interviewScore with an empty value	Only write interviewScore for the evaluate_interview action
Daily Digest subject begins with "==" so a stray "=" appears	Change the subject expression prefix to a single "="
Dashboard actions are plain GET links without authentication	Add a secret token or switch to POST requests